ME5311 Project Report: POD-Based Comparison of Hankel-DMD, SINDy, and NVAR

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Prediction of high-dimensional spatio-temporal systems from measurement data is addressed here with three data-driven methods that span complementary modelling paradigms: Hankel-extended Dynamic Mode Decomposition (Hankel-DMD), Sparse Identification of Nonlinear Dynamics (SINDy), and Nonlinear Vector AutoRegression (NVAR). The dataset comprises $15{,}000$ snapshots of a two-component velocity field on a 64×64 grid with periodic boundary conditions. Proper Orthogonal Decomposition (POD) reduces the $8{,}192$-dimensional state to $r = 31$ modes retaining 92.26% of the training energy, enabling tractable model fitting. At a four-time-unit forecast horizon ($h = 20$ steps), all three methods achieve $R^2 > 0.97$. At a forty-time-unit horizon ($h = 200$), SINDy reaches $R^2 = 0.656$, Hankel-DMD $R^2 = 0.572$, and NVAR $R^2 = 0.464$. Hankel-DMD yields the longest valid prediction time ($\mathrm{VPT} = 5.6$ t.u.) and the best long-horizon accuracy ($R^2 = 0.150$ at $h = 2000$ steps), at the cost of higher computational overhead (1.82 s training). SINDy is the fastest method (< 0.01 s) and best at medium horizons by confining its library to the four dominant POD modes. All methods ultimately diverge at long horizons, consistent with the chaotic nature of the system.

Dynamic Mode Decomposition | SINDy | NVAR | POD | Spatio-temporal prediction

Introduction

Forecasting the future states of high-dimensional spatio-temporal systems from data is a central problem in computational mechanics and fluid dynamics (1). Within the inverse-problem framework of (2), the objective is to recover the dynamical operator $\mathbf{m}$ satisfying $\mathbf{d} = g(\mathbf{m})$, where $\mathbf{d}$ denotes observed future states and g is the forward propagation map. Recovering $\mathbf{m}$ from a measured trajectory and then using $g(\mathbf{m})$ for prediction is precisely the inverse task.

Three paradigms span the interpretability–flexibility spectrum (2). Dynamic Mode Decomposition (DMD) (3, 4) solves the linearised inverse problem for a best-fit linear propagator; extending DMD with Hankel delay-coordinate embedding (5, 6) expands the effective state space and allows the linear model to capture a broader class of nonlinear dynamics via the Koopman operator framework (7, 8). SINDy (9) identifies a parsimonious governing equation from a polynomial candidate library via sparse regression (10), connecting directly to the ℓ_p-norm cost-function formulations discussed in the course. Nonlinear Vector AutoRegression (NVAR) (11) combines delay-coordinate embedding with nonlinear polynomial features and a closed-form ridge-regression readout, offering a computationally efficient reservoir-computing-inspired alternative to gradient-based neural networks (12).

Direct application of these methods to the raw $\mathcal{O}(10^4)$-dimensional snapshots is computationally intractable. Proper Orthogonal Decomposition (POD) (13), implemented via the snapshot method of Sirovich (14), first projects the data onto an energetically dominant low-dimensional subspace, reducing the prediction problem to one over a compact set of temporal coefficients. The combination of POD with data-driven dynamical models has proven effective for reduced-order modelling of complex flows (2, 15). This paper implements and compares all three approaches on a controlled benchmark, evaluates multi-horizon prediction accuracy, and discusses the trade-offs among accuracy, computational cost, and interpretability.

Methods

Data and dimensionality reduction. The dataset contains $N_t = 15{,}000$ snapshots of a two-component planar velocity field on a uniform 64×64 Cartesian grid with periodic boundary conditions ($\Delta t = 0.2$, $T = 3000$). The data are split *chronologically* into training (60%), validation (20%), and test (20%) sets to prevent temporal data leakage. Each snapshot is vectorised to $\mathbf{x}(t) \in \mathbb{R}^n$, $n = 8{,}192$. The mean-centred training snapshot matrix $\mathbf{X} \in \mathbb{R}^{n \times N_{\text{train}}}$ is factorised as $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$; the leading $r = 31$ left singular vectors form the POD basis $\mathbf{U}_r$, capturing 92.26% of the training energy (Fig. 1a). The temporal coefficients $\mathbf{a}(t) = \mathbf{U}_r^\top \mathbf{x}(t) \in \mathbb{R}^{31}$ serve as the state vector for all downstream models.

Hankel-Extended Dynamic Mode Decomposition. To expand the effective state space, a delay-coordinate embedding of depth d is formed:

$$\mathbf{z}(t) = \left[\mathbf{a}(t)^\top, \mathbf{a}(t-1)^\top, \ldots, \mathbf{a}(t-d+1)^\top\right]^\top \in \mathbb{R}^{rd}, \qquad [1]$$

yielding Hankel snapshot matrices $\mathbf{Z}$ and $\mathbf{Z}'$ (shifted by one step). DMD then identifies the linear map $\mathbf{Z}' \approx \mathbf{A}\,\mathbf{Z}$, solved via the reduced propagator $\widetilde{\mathbf{A}} = \mathbf{U}_r^\top \mathbf{Z}' \mathbf{V} \mathbf{\Sigma}^{-1}$ (4, 5). The delay depth $d \in \{12, 16, 20\}$ and eigenvalue damping $\rho \in \{0.995, 0.999\}$ are selected on the validation set; the best configuration ($d = 20$, $\rho = 0.999$) gives a state dimension of $rd = 620$. Eigenvalues with magnitude exceeding unity are stabilised to $|\lambda| \to \rho$. Predictions are generated by iterating the Hankel propagator and extracting the leading r components of $\mathbf{z}$.

Sparse Identification of Nonlinear Dynamics. SINDy (9) posits that the POD dynamics satisfy $\dot{\mathbf{a}} = \mathbf{\Xi}\,\mathbf{\Theta}(\mathbf{a})^\top$, where $\mathbf{\Theta} \in \mathbb{R}^{N \times p}$ is a polynomial candidate library and $\mathbf{\Xi}$ is a sparse coefficient matrix. Because higher POD modes (indices 5–31) carry $< 1\%$ energy each and evolve nearly stochastically, SINDy is applied only to the $r_s = 4$ energetically dominant modes, retaining 78.5% of training energy. The identification problem is formulated as

$$\mathbf{\Xi}^* = \arg\min_{\mathbf{\Xi}} \left\| \dot{\mathbf{A}}_s - \mathbf{\Xi}\,\mathbf{\Theta}^\top \right\|_F^2 + \alpha \|\mathbf{\Xi}\|_F^2, \qquad [2]$$

and solved via the Sequential Thresholded Least Squares algorithm ($\tau = 0.005$, $\alpha = 0.1$), using a degree-1 library. The remaining $r - r_s$ coefficients are held fixed at their initial test values during autoregressive rollout. The prediction mode and amplitude-shrink factor ($\alpha_s = 0.9$) are selected on the validation set.

Nonlinear Vector AutoRegression. Following (11), NVAR forms the same delay-coordinate vector $\mathbf{z}(t) \in \mathbb{R}^{rd}$ as Eq. (1) and maps it through a polynomial feature function $\mathbf{\Theta} : \mathbb{R}^{rd} \to \mathbb{R}^p$. The readout matrix is obtained in closed form via ridge regression:

$$\mathbf{W}^* = \arg\min_{\mathbf{W}} \left\| \mathbf{Y} - \mathbf{\Theta}(\mathbf{Z})\mathbf{W} \right\|_F^2 + \alpha \|\mathbf{W}\|_F^2, \quad \alpha = 10^{-2}, \qquad [3]$$

where $\mathbf{Y}$ are the one-step-ahead targets. Hyperparameters (delay $d \in \{2, 4, 8\}$, polynomial degree, ridge α) are selected on the validation set; the best configuration ($d = 8$, degree 1, $\alpha = 10^{-2}$) renders NVAR a regularised linear regression in the $rd = 248$-dimensional delay space. Predictions are generated autoregressively.

Evaluation metrics. Prediction quality is assessed at multiple forecast horizons h using the following metrics: the coefficient of determination R^2; root mean squared error (RMSE) and its normalised variant (NRMSE $= \text{RMSE}/\sigma_{\text{true}}$); Pearson correlation; and Valid Prediction Time (VPT), defined as the first horizon at which NRMSE exceeds 0.3.

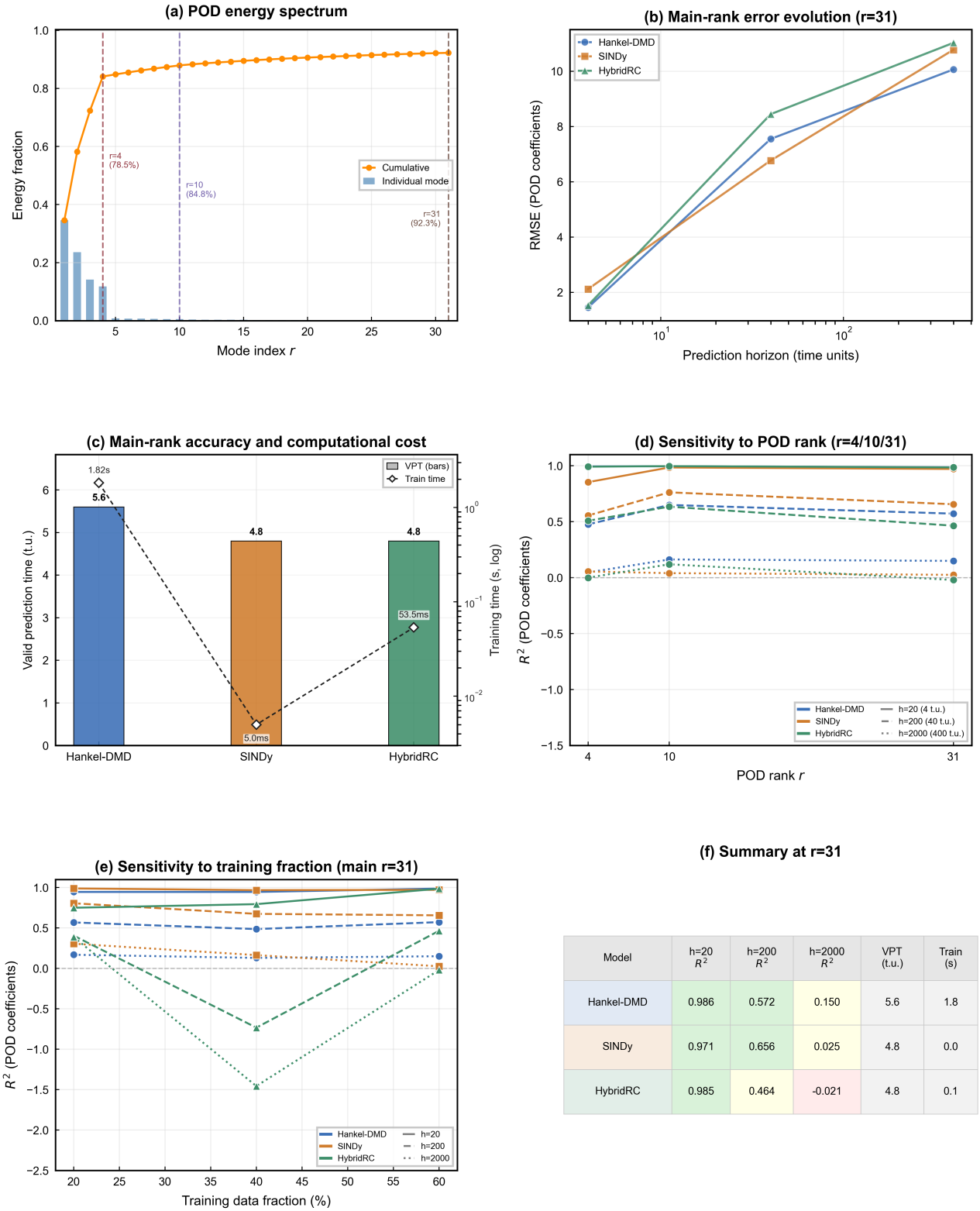
Results and Discussion

Figure 1 summarises the prediction results across all methods. The POD energy spectrum (panel a) reveals a sharp decline from mode 4 (11.8%) to mode 5 (0.73%), consistent with four energetically dominant mode pairs imposed by the prescribed external forcing wavenumber; the chosen rank $r = 31$ retains these dominant modes together with all weaker residual modes up to the 92% energy threshold.

Prediction accuracy. At $h = 20$ steps ($t = 4$ t.u.), all three methods achieve $R^2 > 0.97$ and correlation > 0.99 (panels c, f), with Hankel-DMD performing best ($R^2 = 0.986$), followed by NVAR ($R^2 = 0.985$) and SINDy ($R^2 = 0.971$). Performance differences become pronounced at longer horizons: at $h = 200$ ($t = 40$ t.u.), SINDy leads with $R^2 = 0.656$, followed by Hankel-DMD ($R^2 = 0.572$) and NVAR ($R^2 = 0.464$). SINDy's medium-horizon advantage arises because operating on the $r_s = 4$ dominant modes avoids contamination from the small-scale stochastic fluctuations present in modes 5–31, which degrade the rollout of full-state models. At $h = 2000$ ($t = 400$ t.u.), Hankel-DMD maintains $R^2 = 0.150$ while SINDy degrades to $R^2 = 0.025$ and NVAR to $R^2 = -0.021$. Hankel-DMD also achieves the longest valid prediction time (VPT = 5.6 t.u.), while SINDy and NVAR share VPT = 4.8 t.u. All methods eventually diverge at long horizons (panel b), consistent with exponential error growth from positive Lyapunov exponents in chaotic systems (16). Rolling-origin evaluation confirms these trends: at $h = 200$, Hankel-DMD yields $R^2 = 0.447 \pm 0.164$ and NVAR $R^2 = 0.398 \pm 0.197$, while SINDy shows higher variance ($R^2 = 0.270 \pm 0.662$) due to sensitivity to initial conditions when operating in the reduced subspace.

Computational cost and interpretability. SINDy trains in < 0.01 s and yields explicit governing equations for the four dominant modes, making it the most interpretable and fastest method; its medium-horizon accuracy demonstrates that focusing on the energetically dominant subspace is beneficial. NVAR requires $\sim$0.05 s for the closed-form ridge solve, with no iterative optimisation; its degree-1 delay library renders it a linear model in the augmented state space, analogous to Hankel-DMD but without the spectral decomposition step. Hankel-DMD requires $\sim$1.82 s training due to the large state dimension ($rd = 620$), but achieves the best long-horizon accuracy and VPT by exploiting the full delay-coordinate structure to approximate Koopman eigenfunctions (7, 8). Overall, the three methods offer complementary trade-offs: SINDy for interpretability and medium-horizon accuracy, NVAR for computational efficiency, and Hankel-DMD for long-horizon robustness; all are ultimately bounded by the fundamental predictability limit of chaotic dynamics.

AI Declaration. AI-assisted language tools (Claude Sonnet 4.6) were used for limited drafting and wording refinement of the manuscript; all analysis design, code implementation, result verification, and technical interpretation were completed by the author.



Model	h=20 R^2	h=200 R^2	h=2000 R^2	VPT (t.u.)	Train (s)
Hankel-DMD	0.986	0.572	0.150	5.6	1.8
SINDy	0.971	0.656	0.025	4.8	0.0
HybridRC	0.985	0.464	-0.021	4.8	0.1

Fig. 1. Prediction analysis for Hankel-DMD, SINDy, and NVAR. (a) POD energy spectrum on training data; bar: per-mode fraction; curve: cumulative energy; vertical dash-dot: $r = 31$; horizontal dashed: 90% threshold. (b) RMSE of predicted POD coefficients versus forecast horizon. (c) True and predicted trajectories of the leading POD coefficient $a_1(t)$ over the first 200 test steps. (d) Reconstructed velocity-magnitude snapshots at test step $t = 50\Delta t$: reference (top), Hankel-DMD and SINDy predictions (bottom row). (e) Pearson correlation versus forecast horizon. (f) R^2 comparison at forecast horizons $h \in \{20, 200, 2000\}$ steps for all three models.

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